ATROPINE

Organophosphate antidote

Anticholinergic toxicity

CHEMICAL NAME (1R,5S)-8-methyl-8-azabicyclo[3.2.1]octan-3-yl (2RS)-3-hydroxy-2-phenylpropanoate (sulfate)

CLASS Antimuscarinic (anticholinergic) agent

TRADE NAMES Atropine sulfate injection · AtroPen · Isopto Atropine (eye drops)

THERAPEUTIC USES / INDICATIONS

- Symptomatic bradycardia (first-line drug).
- Organophosphate and carbamate poisoning (large doses).
- Reduce secretions and vagal reflexes in anaesthesia; with neostigmine for reversal.
- Mydriasis/cycloplegia (eye drops); uveitis.

PHARMACODYNAMICS

- Competitive antagonist at **muscarinic receptors** → blocks vagal effects.
- ↑ HR and AV conduction; ↓ salivary, bronchial and sweat secretions; mydriasis; ↓ GI and bladder motility.
- Does not reverse nicotinic effects (weakness) of organophosphates — use pralidoxime.

ADMINISTRATION

- **IV** bolus (fast); **IM**, **SC**, **endotracheal** (2–3× dose).
- Organophosphate poisoning: give IV and double the dose every 5 min until atropinised.
- Eye drops: press on inner corner of eye 1–2 min after instillation.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dry mouth, blurred vision, photophobia.
- Tachycardia, flushing.
- Urinary retention, constipation.

ADVERSE EFFECTS (SERIOUS)

- Anticholinergic toxicity: hot, dry, red skin, dilated pupils, delirium, hyperthermia ("hot as a hare, dry as a bone, red as a beet, blind as a bat, mad as a hatter").
- Paradoxical bradycardia with doses <500 mcg.
- Angle-closure glaucoma.
- Ventricular arrhythmias.

DOSAGES

- **Bradycardia (adult):** 500 mcg IV, repeat every 3–5 min to max 3 mg.
- **Organophosphate:** 1–2 mg IV (child 0.02 mg/kg), doubling every 5 min until secretions dry, HR >80, SBP >80, clear chest; then infusion 10–20% of loading dose/h.
- **Premedication:** 300–600 mcg IV/IM.
- **Children (bradycardia):** 20 mcg/kg (max 600 mcg).

CONTRAINDICATIONS & CAUTIONS

- Angle-closure glaucoma (relative).
- Myasthenia gravis (except with anticholinesterase).
- Paralytic ileus, pyloric stenosis, toxic megacolon; prostatic obstruction.
- No absolute contraindication in life-threatening bradycardia or poisoning.

KEY INTERACTIONS

- **Other anticholinergics** (TCAs, antihistamines, antipsychotics): additive toxicity.
- **Metoclopramide, domperidone:** effects opposed.

PHARMACOKINETICS

- **Onset:** IV 1 min; IM 15 min. **Duration:** 2–4 h (eye effect 7–12 days).
- Crosses blood–brain barrier and placenta.
- Metabolised in liver; excreted in urine. **t½:** 2–4 h.

NURSING MANAGEMENT

- Continuous ECG during IV use; record HR response.
- Organophosphate: assess atropinisation endpoints frequently (chest auscultation, HR, BP, pupils, secretions).
- Monitor temperature, urine output (retention), bowel sounds.
- Watch for confusion/agitation (toxicity).
- Eye drops: warn about blurred vision.

PATIENT EDUCATION

- Blurred vision and light sensitivity after eye drops may last days; avoid driving.
- Report difficulty passing urine, confusion, very fast heartbeat.

KEY POINTS

- Muscarinic antagonist.
- Organophosphates: double dose until secretions dry.
- Bradycardia: 500 mcg IV, max 3 mg.
- Toxicity: hot, dry, red, mydriasis, delirium.

SOURCES CHECKED **[386]** Atropine sulfate 0.5 mg/5 ml, solution for injection in pre-filled syringe (UK SmPC) **[387]** Atropine Sulfate 600 micrograms/ml solution for injection (UK SmPC) **[388]** ATROPEN Auto-Injector, IM (US label)

Full references p. 432–457